

2(a)	to, absorb / remove / take up, the carbon dioxide produced (during respiration of blowfly larvae) ;	1
2(b)	<i>direction</i> the coloured fluid will, move upwards and <i>explanation</i> (oxygen is, taken in / used in respiration) so, decreases the, volume / pressure of gas (in test-tube) ;	1

Question	Answer	Marks
2(c)	1 remove substance X ; 2 replace X with an inert material / glass beads / AW ;	2
2(d)	<i>any one from:</i> temperature ; number of, (blowfly) larvae ; mass of, (blowfly) larvae ; age of, (blowfly) larvae ; food source for, (blowfly) larvae ; mass / number of, substance X / inert material / beads ; time to, equilibrate ; time, for measurement (of meniscus movement) ;	1
2(e)	<i>any two from:</i> 1 the larvae were not fully adjusted / equilibrated ; 2 the larvae were at a lower temperature (than the apparatus) ; 3 the larvae were less active ;	2
2(f)	5.34 ;; <i>incorrect answer max one for working:</i> $(\pi \times 0.5^2 \times 68) \div 10$;	2

Question	Answer	Marks
2(g)	0.56 ;; <i>incorrect answer max one for working:</i> 3.00 / 5.34 ;	2